



Ahmedabad
University

CSE623 : Machine Learning

**Machine Learning-based Estimation of Muscle
Cross-Sectional Area (ACSA) from Segmented
Ultrasound Images of Lower Limb Muscles for
Sports Performance Analysis**

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Abstract

This study entails a machine learning pipeline of automatic segmentation of the Rectus Femoris (RF) muscle on ultrasound images and consequent Anatomical Cross-Sectional Area (ACSA) estimation. Images and their respective ground-truth masks were obtained at pixel level and preprocessing techniques like region extraction, median filtering and CLAHE was performed to optimize image quality and minimize noise. A pool of 24 pixel-wise features of intensity, texture, gradient and spatial were obtained. A Random Forest classifier was trained with a hard-negative mining strategy to enhance boundary learning, and morphological post-processing was done to hone the result of segmentation. It was found that the model used performed well according to Dice coefficient and Intersection-over-Union (IoU). Lastly, ACSA was determined by converting the pixels which were segmented into physical area using calibration factors. Future improvement will involve better feature representation, better model parameter optimization and better post-processing to deliver better and more accurate segmentation in challenging ultrasound conditions.

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1 What We Have Done

- We loaded ultrasound images of the Rectus Femoris (RF) muscle and their respective ground-truth segmentation masks. The images were manipulated in their original resolution and made matrices to allow being manipulated at the pixel level. The masks were translated into binary forms to become able to distinguish between muscle and background regions.
- We used preprocessing techniques to enhance the quality of images and noise reduction. These involved thresholding to extract the valid scan region, cropping to the region of interest, median filtering to remove speckle noise and Contrast Limited Adaptive Histogram Equalization (CLAHE) to increase contrast.
- We obtained a wide set of 24 pixel-wise features that reflected intensity, texture, gradient, and spatial features. These were multi-scale mean and standard deviation, entropy, variance, Laplacian response, directional gradients, texture features based on GLCM, and spatial priors like distance to the boundary and centroid.
- To build the training data set, we employed hard-negative mining approach where we sampled 3000 pixels each image, comprising of muscle pixels, boundary negatives and background pixels. This contributed to enhancing the difficult boundary region learning of the model and augmenting segmentation strength.
- We used a Random Forest classifier on the extracted features to predict the type of pixel, muscle or background (1 or 0). Morphological processes like closing, hole filling and removal of small connected components were used to refine the predicted segmentation masks.
- We compared the segmentation of performance on the use of Dice coefficient and Intersection-over-Union (IoU), with good accuracy in validation and test databases.
- Lastly, we calculated the Anatomical Cross-Sectional Area (ACSA) by transforming the number of pixels that have been segmented into physical area using a known pixel-to-area constant, which allows estimation of muscle area automatically using ultrasound imaging.

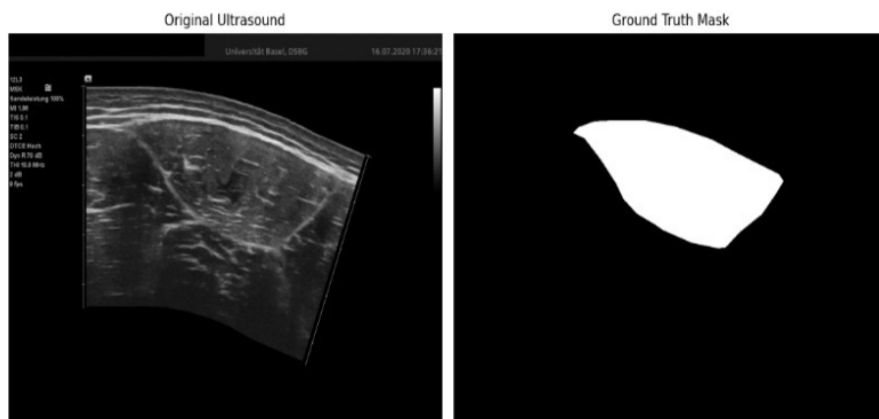


Figure 1: Ultrasound image & Ground Truth Segmentation Mask

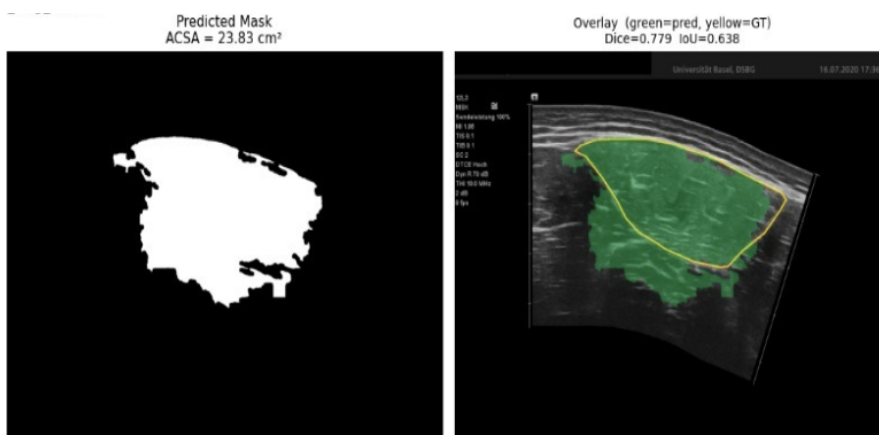


Figure 2: ASCA value & Overlay

2 What We Aspire to Do

- More work will be done to improve the accuracy of segmentation of the Random Forest model. To improve the feature set, we intend to add more discriminative texture and gradient-based features and to obtain improved multi-scale representations to enhance boundary detection in low-contrast, noisy ultrasound images.
- We also want to tune the best combination of hyperparameters of the Random Forest classifier to the number of trees, maximum depth, and the method of feature selection at each split. Further, better sampling will be investigated in order to capture the problematic boundary areas and minimize class imbalance.
- In order to further improve the segmentation output, we will explore better methods of post-processing, including adaptive morphological operations and contour-based refinement to achieve smoother and better quality muscle boundaries.
- Finally, we will enhance the ACSA estimation with a better calibration process and examination of the correlation between the accuracy of segmentation and area estimation with more accurate and consistent results.

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